

Financial development as a new determinant of energy diversification: The role of natural capital and structural changes in Australia

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ABSTRACT

This paper investigates the relationship between financial development and energy diversification for Australian economy by considering natural capital, structural changes, economic growth, and export diversification as additional determinants of energy diversification demand function. The bootstrapping ARDL cointegration approach is applied for examining cointegration relationship between energy diversification and its determinants by considering presence of structural break stems in the series. Our empirical analysis reveals that financial development affects energy diversification positively. Natural capital is positively linked with energy diversification. Structural changes have favourable effect on energy diversification. Economic growth declines energy diversification. The relationship between export diversification and energy diversification is negative. Financial development and energy diversification have a nonlinear relationship between with an inverted-U shaped. The bootstrapping ARDL Granger causality analysis reveals the presence of feedback effect between financial development and energy diversification. This empirical analysis provides new policy insights for Australian policy makers using financial development as economic tool to boost up energy diversification.

1. Introduction

There is a constant effort on the part of every government to improve their nation's energy diversification to reduce dependency on fossil fuels. Energy diversification has historically provided numerous advantages from an economic perspective. It is in light of this, that policy makers look for economic factors which can propel energy diversification action to a higher trajectory. Financial development is seen as an important agent in various economic developmental activities. For instance, [Pei et al. \(2023\)](#) noted that financial development benefits the economy by reducing CO₂ emissions. Financial development has also leads to the conservation of scarce mineral resources ([Nassani et al., 2021](#)), thereby promoting the environment and natural resource conservation. [Nibedita and Irfan \(2023\)](#) discover that financial development moderates carbon emissions caused by energy diversification in a sample seven emerging economies. These studies have been able to

measure economic and environmental benefits of financial development. This essentially leads us to assume that there is a profound need on the part of policymakers to recognize the role of financial development in economic policy. One important direction is the act of developing a wise-mix of financial and energy related policy measures for advancing the economic agenda of the nations ([Wang et al., 2020](#)). This emphasises the importance of investigating the relationship between financial development and energy diversification in order to devise policy initiatives.

In the seminal work by [Wrugler \(2000\)](#) it has been highlighted that one of the most significant advantages of financial development in the long run is that it helps divert investments in modern technologies and curbs the flow of fund in lesser productive sectors. This leads to a significant impact on the reduction of emissions. Some recent works have also studied this aspect and agree with this results. For instance the study by [Nibedita and Irfan \(2023\)](#) established that energy diversification

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assisted by financial diversification has a favourable impact on carbon emissions reduction when they studied the seven emerging economies of the world. They established this development in the long run. On the similar footing, other study (Zhang et al., 2021) also indicates that financial development and renewable energy consumption are inter-linked and collectively have a positive impact on reducing carbon emissions in the short run. This is supported by the other studies which demonstrated the cointegration between financial development and renewable energy consumption in the context of ASEAN nations (Hendrawaty et al., 2022).

Natural capital constitutes the living and non-living elements that contributes to the creation of products and services consumed by people (Guerry et al., 2015). Natural capital is a vital component for carrying out economic activity. Nevertheless, in an increasingly urbanised world where people live in high rise- buildings and consume products manufactured in distant parts of the world, the importance of natural capital seems to have diminished. But, it is the very source, invisible to consumer that aids in the manufacturing of goods and creation of services. The tremendous increase in population and the associated exponential rise in demand for goods and services have resulted in the rapid depletion of natural capital (Polasky and Daily, 2021). Natural capital that has faced fast decline includes forests, wetlands, minerals, coal reserves, natural cleansing of environmental pollutants, and so on, which will have adverse consequences for economic development. While certain kinds of natural capital such as coal, oil, and gas, are easily quantified, the value of other forms of natural capital such as process of pollination, water flows, nutrients conversion, control of pathogens are never directly measured and hence less appreciated and understood. The significant role of natural capital in serving the needs of humanity has now started attracting the attention of modern economists. This has resulted in discovery of the ecosystems that aid and assist in service delivery, one notable break-through in this direction was the publication of the book (Millennium Ecosystem Assessment, 2005), which highlighted the importance of the ecosystem in delivery of business services. Concerns for conservation of natural capital have thus emerged as a significant priority for the policy makers worldwide. A significant aspect of ecosystem conservation is the reduction of emissions, which effectively means that ecosystem conservation is linked to the control of emissions through energy diversification (Islam et al., 2022). As a result, it is important to evaluate the relationship between natural capital and energy diversification. This research attempts to assess the linkage of natural capital with energy diversification to better explain their nexus. Structure change refers to the movement of labour and other productive resources from low-productivity to high-productivity economic activity. The level of structural change is determined by the complexity and diversity of exports, the intensity of capital investment, and the percentage of industry and services to total GDP that these elements now account for. Structure change is also possible within an industry, but only if the industry's limitations are recognised and successfully overcome. Higher economic growth is achieved when the factors of production move from the low productive activity to higher ones, resulting in what is termed as 'structure dividend' (Zhou et al., 2021). For many decades, economists ignored structural change since it was not viewed as crucial in explaining economic progress. Multiple recent studies have demonstrated the effective role of structural changes in demonstrating economic growth (Li et al., 2021a, 2021b, Valadkhani and O'Mahony, 2018).

Additionally, energy diversification leads to exploration of a range of energy sources to meet energy demand of a country. Although energy diversification has been studied in the past several studies (Ahmed et al., 2022; Akrofi, 2021; Gozgor and Paramati, 2022; Islam et al., 2022), but none have attempted to analyse the role of financial development in energy diversification. The role of financial development has also been explored to explain different aspects of energy consumption such as renewable energy (Lahiani et al., 2021); energy innovation (Baloch et al., 2021) energy consumption (Hendrawaty et al., 2022); energy

utilization (Yang et al., 2021), but none of the studies have explored the impact of financial development on energy diversification. This study therefore, occupies this place in the current body of knowledge to investigate the energy diversification. This study therefore is an endeavour to explore and explain the role of financial development as a significant indicator for energy diversification. The Australian economy has been steadily growing, real GDP rates projected to reach 1.9% in 2023 and 1.6% in 2024, respectively. The Reserve Bank of Australia has quickly increased interest rates in response to high inflation that is eroding family purchasing power. The labour market's tightness is anticipated to ease as GDP slows. When the labour market cools and supply chain bottlenecks loosen, inflationary pressures will lessen (OECD, 2022). After having dealt with the effects of pandemic for two fiscal years, Australia is positioned well to progress to a higher economic growth. Furthermore, the Australian economy is witnessing structural change over the past two decades. The Australian economy has gradually moved away from manufacturing and agriculture towards services, with the mining sector recently attaining an increased significance (Adeney, 2018). Additionally, the economic growth in Australia has been on a consistent growth trajectory and studies reveal that it has registered a sustained economic growth for the past 27 years (Leal et al., 2019). Yet, the same study reveals, Australia is heavily reliant on the non-renewable energy sources. In fact, Australia is one of the top 10 CO₂ emitter countries (Adebayo and Acheampong, 2022). In addition, Australia also exports a large chunk of its natural resources, especially coal to several countries. In the year 2011, Australia was ranked the largest exporter of coal in the world (Leal et al., 2019), indicating its reliance on non-renewable energy sources for exports income. Coal, Coke and gas still continue to dominate Australia's exports, accounting for 22% of total exports. This indicates that export diversification, notably, exports of fossil based fuel need to be reduced if, it has to create a sustainable export model.

An assessment of Australia's economic policies at this juncture will help improve developmental initiatives and increase its economic resilience. As a result, this study acquires crucial relevance, since it is critical from the perspective of sustainability to identify the components that will aid Australia in strengthening its energy diversification strategy. Moreover, the factors being considered for the purpose of this study have not been investigated in the past. This study contributes to existing literature considerably. (i), We use financial development, natural capital, structural changes, economic growth and export diversification for measuring their impact on energy diversification. (ii), We use traditional and advanced unit root test containing information of single unknown structural break stems in the series. (iii), The bootstrapping ARDL cointegration approach for examining cointegration relationship between energy diversification and its determinants by considering presence of structural break stems in the series. (iv), The bootstrapped ARDL based causality approach is applied examining causal relationship between the variables in the presence of structural break in the series. Our findings confirm the presence of cointegration between financial development and energy diversification. Further, financial development stimulates energy diversification. Natural capital positively affects energy diversification. Structural changes also encourage energy diversification. Economic growth is negatively linked energy diversification. An inverted-U shaped relationship exists between financial development and energy diversification. The bootstrapping ARDL Granger causality analysis verifies the feedback effect between financial development and energy diversification.

The structure of this paper is organized as following: Section-II presents the literature review. Section-III builds the theoretical background and empirical model. Section-IV presents the empirical results and empirical results and findings are discussed in Section-V. Section-VI presents conclusion, policy implications, limitations and directions for future research.

2. Literature review

Energy diversification involves incorporating new energy sources into the mix, in order to lessen the reliance on a single energy source. It is described as increasing the share of multiple renewable energy sources (Stirling, 2010). Both developing and developed nations are making constant efforts to diversify their energy sources. The diversity of energy is essential from the standpoint of avoiding the risks associated with external shocks and to make a gradual shift to the sustainable sources of energy (Gözgör and Can, 2017). Energy diversification not only provides a safeguard against external supply risks, but also ensures steady economic activity during the variation in internal energy production. This process of diversification of energy sources is not just confined to the nations which have external dependencies on the supply of energy, but also those nations which export energy (Albassam, 2015). Energy diversification offers a variety of benefits to the economy, like adding to the technological advancement of the nation, reduction in emissions, and saving foreign currency reserves by reducing the import of energy, finally leading to sustainable economic growth (Kireyev, 2021; Mania and Rieber, 2019). According to the reports (de Atholia et al., 2020), Australia will continue its energy diversification by increasing investments in sustainable energy sources. As a result of such investments, Australia has noticed a steady decline in non-renewable energy sources while increasing its share of renewable sources such as solar, wind, and hydro energy. Financial investments have substantially increased in the energy sector. Studies also prove that financial development leads to the adoption of cleaner sources of energy (Yao and Zhang, 2021). This provides acceptance to the hypothesis that financial development leads to improvement in energy diversification.

Qamruzzaman (2022) found that in the context of BRICS nations, there was a directional relationship between energy diversification and financial development explained through the flow of foreign capital. Although this direction varied between the nations under study. The study also found that the financial development has a significant and positive association with the energy diversification in the context of developing nations. Moreover the study by Udemba et al. (2023) established a strong impact of financial development on the carbon emissions via energy diversification efforts in the context of China. This establishes the entrenched association between financial development and energy diversification. Moreover, some more evidences are available to further reinforce the association between financial development and energy diversification (Raihan and Voumik, 2022). They suggest that there is both long and short-run influence of financial development on carbon emissions propelled by energy diversification among over factors.

Energy diversification can be impacted in several ways. For instance, the World Bank (2021) report on the wealth of nations has specified that natural capital as one form of capital which is essential to be accounted for. Considering the significant role of natural capital in the process of creation of products and services, these assets are of greater importance as these last for a longer duration and are considered as wealth of the nation (Hamilton and Hepburn, 2017). Australia is a wealthy country with abundant natural resources and a sizable landmass. Almost a century ago, it liberalised trade, as sustainable growth needed open immigration policies. Australia has benefited from globalisation during the past forty years thanks to lower trade barriers, smoother money flows, the diffusion of technology, free movement of labour, and effective resource utilization (Adebayo and Acheampong, 2022). The contribution of natural capital has been studied in the context of Scotland for understanding the role of wind energy installations as a mitigation measure for CO₂ emissions (Shepherd et al., 2021). Studies have also indicated how mismanagement of natural resources has resulted in the erosion of the value of natural capital and also identified the specific losses which were incurred (Tolessa et al., 2017). Authors have also proposed a new framework for presenting the linkage between ecological perspective and the economic consequences (Bateman and Mace,

2020). It therefore demonstrates that natural capital can be deterministic of energy diversification in an economy.

Structural changes denote transformation in the structure of an economy, which may occur as a result of any shift in a few or multiple components of the economy. The structural changes can occur as a result of multiple drivers, either large or minor in nature (van Neuss, 2019). Studies have used structural change as a measure or as an outcome of multiple events. For instance, studies have measured the impact of digitalization (Matthess and Kunkel, 2020), technological innovation (Xu et al., 2022), etc. on the structural changes. Some studies use role of agriculture (Zhang and Diao, 2020) to measure its impact on structural changes while others use socio-economic and climate change mitigation situations as a result of changes (Lefèvre et al., 2022). This indicates that structural changes are vital economic dimension that has been successfully employed by the researchers to explain the economic status of countries (Erumban et al., 2019). This study therefore, include structural changes as a variable of study to explain its role in energy diversification.

Economic growth has been linked to the energy mix in multiple studies in the literature. For example, studies have explained the policy measures to ensure sustainable economic growth in the post-pandemic scenario, through energy diversification (Bashir et al., 2022). The inclusion of fossil fuel in the energy mix and its impact on economic growth has also been studied in the past (Kibria et al., 2019). Energy diversification is perhaps one of the leading reasons for the Paris agreement where energy diversification or shifting to non-fossil sources was targeted by 192 nations to promote sustainable economic growth (Lovell et al., 2022). Very recent studies have indicated that economic growth can be augmented if the nations shift to renewable sources of energy via energy diversification efforts (Díaz et al., 2019). The empirical evidence provided by Rajbhandari and Zhang (2018) proved that energy efficiency can lead to higher growth in economic development. This indicates that there is a plethora of evidence in the literature to prove the linkage between energy diversification and economic growth.

Export diversification is an economically beneficial exercise which helps nations to cut down large dependencies on specific exports and provides greater alternatives for generating exports revenues. Needless to mention here are the advantages of export diversification. Studies have proved that the export diversification has led to environmental degradation due to higher dependence on fossil fuels (Can et al., 2021). Energy consumption as well as export diversification has also been empirically found to have a strong linkage (Shahzad et al., 2021). The linkage between renewable energy sources and export diversification has also been established in the study by Li et al. (2021a, 2021b). However, there are contrary findings reported by Fatima et al. (2022) that studied the impact of export diversification in the 6 Gulf Cooperation Council (GCC) countries. Their study revealed that there is an inverse association between export diversification and consumption of renewable energy. This helps us to understand energy diversification and export diversification interdependence, enabling us to move forward and assess this association in this study.

Existing literature has used factors for determining various dimensions and outcomes, but none of the studies have examined the factors considered in this study. It is in light of the above reviewed literature; this study is placed on a unique ground to explore the determinants of energy diversification.

3. Theoretical background, empirical modelling and data

The rapid change in technology has made it easier to generate energy from multiple sources, creating opportunities for forming energy-mix. This process of creating a diversity of sources for the fulfilment of energy requirements must be coupled with effective policy measures. The energy concentration causes ecological footprint to increase, whereas energy diversification has a long-term effect of reducing ecological

footprint, supporting sustainability efforts (Kouton et al., 2022). There are various theoretical approaches which provide an explanation of how energy diversification can be achieved in an economy. The energy ladder approach by Hosier and Dowd (1987) assumes, that with the rise in household income, energy consumers will gradually move away from traditional energy to technologically advanced energy sources, leading to energy diversification. The next prominent theoretical approach which improves the energy ladder hypothesis (Harrington et al., 2020) postulates that, although the rising income of households does result in energy diversification, it does not cause major transition from one form of energy to another. According to the growth hypothesis, using alternative energy sources will accelerate the pace of economic progress (Ahmed et al., 2022). Another prominent model which explains the energy diversity is the Jevons' paradox, which explains that even with an expanded mix of energy sources, the efficiency will not improve, as a shift from one energy source to another will only shift the demand from one to another. It is evident from above that the theoretical approaches in the recent literature have some theoretical frameworks that attempt to explain the dimensions of energy diversification. This study adopts a novel approach and examines financial development as a determinant of energy diversification. Based on the above theoretical background, the general form of energy diversification demand function can be modelled as given below:

$$E_t = f(F_t, N_t, S_t, Y_t, X_t) \quad (1)$$

Following Shahbaz et al. (2020a, 2020b, 2022), we convert all the variables into natural-log. Natural-log conversion may help to smoothen the sharpness stems in time series data. The coefficients we obtain from a natural-log scale are directly interpreted with proportional differences and such coefficients are understood as elasticities. In such a scenario, we use the log-linear specification of the energy diversification demand function rather than the simple linear specification. The empirical equation of the energy diversification demand function is modelled as the following:

$$\ln E_t = \alpha_1 + \alpha_2 \ln F_t + \alpha_3 \ln N_t + \alpha_4 \ln S_t + \alpha_5 \ln Y_t + \alpha_6 \ln X_t + \mu_t \quad (2)$$

where $\ln$, E_t , F_t , N_t , S_t , Y_t and X_t represent natural-log, energy diversification, financial development, natural capital, structural changes, economic growth and export diversification. μ_t is residual term. We assume that $\alpha_2 > 0$ if financial development stimulates energy diversification otherwise $\alpha_2 < 0$. $\alpha_3 > 0$ if natural capital is positively linked with energy diversification otherwise $\alpha_3 < 0$. If structural changes positively affect energy diversification then $\alpha_4 > 0$ otherwise $\alpha_4 < 0$. $\alpha_5 > 0$ if economic growth favours energy diversification and $\alpha_5 < 0$ if economic growth declines energy diversification. If exports diversification lowers energy diversification then $\alpha_6 < 0$ otherwise $\alpha_6 > 0$. To examine the non-linear relationship between financial development and energy diversification, we added a squared term of financial development to the energy diversification demand function, i.e. Eq. (3). It is assumed that energy diversification is accompanied with financial development at the initial stages of the financial sector and starts to decline after a threshold point as the financial sector develops more. In such a scenario, an extended form of the empirical equation of energy diversification demand function is modelled as given below:

$$\ln E_t = \beta_1 + \beta_2 \ln F_t + \beta_3 \ln F_t^2 + \beta_4 \ln N_t + \beta_5 \ln S_t + \beta_6 \ln Y_t + \beta_7 \ln X_t + \mu_t \quad (3)$$

The relationship between financial development and energy diversification is inverted-U shaped if $\alpha_2 > 0$, $\alpha_2 < 0$. There would be a U-shaped association between financial development and energy diversification if $\alpha_3 < 0$, $\alpha_3 > 0$.

This study employs quarter frequency over the period of 1995–2021 for Australian economy. We collect data for GDP (local currency) and

domestic credit to private sector (as share of GDP) from world development indicators (CD-ROM, 2022). The data on energy diversification is borrowed from Gozgor and Paramati (2022).¹ The data on export diversification is collected from IMF-DFID (<https://www.imf.org/external/np/res/dfidimf/diversification.htm>).² The data on natural capital³ and structural changes⁴ is freely available from United Nations Conference on Trade and Development (<https://unctad.org/topic/least-developed-countries/productive-capacities-index>). We also use total population data to convert data into per capita units except energy diversification, natural capital, structural change and export diversification. We apply quadratic match-sum method to transform annual data into quarter frequency following Helmi et al. (2014a, 2014b), Shahbaz et al. (2020a, 2020b).

4. Methodological framework

4.1. ADF unit root and structural break ADF test

We test the stationarity properties of the variable by considering the traditional ADF unit root test (see, Dickey and Fuller, 1981). However, ADF unit root may produce ambiguous empirical results because it ignores the occurrence of multiple structural breaks in the series (Shahbaz et al., 2018). Kim and Perron (2009) proposed an advanced ADF unit root test to resolve this issue. The incorporation of a single unknown structural break is a very crucial remedy that can produce partial estimates. For instance, the Augmented Dickey-Fuller (ADF) test disregards the nonlinearity and structural discontinuities in the selected series that can be the cause of unit root issues. The empirical unit root tests can accept the null hypothesis, although it should be true, and vice versa, due to their weak level of justification and ambiguous results. In this context, advanced ADF unit root testing is a suitable alternative to regular ADF unit root testing. With just one unknown structural break, the nonlinearity problem is solved, and this provides consistent and reliable empirical results. This study is based on Leybourne et al. (1998a) empirical foundations and computed test statistics (Enders and Lee, 2012), and actually the t -ratio of ψ the ordinary least squares regression:

$$e_t = d(t) + \psi_1 e_{t-1} + u_t \quad (4)$$

$d(t)$ enters in the Eq. (4) as a function of t , and μ_t represents a stationary disruption with variance σ^2 . As a result, it is to be noted that e_t is weakly dependent and its initial value is fixed. Eq. (4) can be used to estimate the values and test the null hypothesis of a unit root, for instance ($\psi_1 = 1$) if the functional form of $d(t)$ is known. Since, we don't know the functional form of $d(t)$, therefore, if $d(t)$ is mis-specification such case it might be difficult to test for ($\psi_1 = 1$). However, it is feasible to approximate and test the hypothesis by applying the Fourier expansion given as:

¹ <https://www.sciencedirect.com/science/article/pii/S0140988322001463>

² The index for exports diversification is generated by IMF staff in a research collaboration named IMF-DFID. The exports diversification index contains three sub-indices such as export diversification index, extensive margin and intensive margin. Lower values of these indices mean high diversification and vice versa.

³ Natural capital comprises of the availability of extractive and agricultural resources. These resources include rents generated from the extraction of natural resource with less cost of extraction for such resources. Natural capital declines as material intensity goes up just to capture commodity dependence.

⁴ Structural change basically shows the transfer of labour and other related productive resources from low productive to high productive economic sectors (activities). This movement factors of production captures by application of sophistication and variety exports, the intensity of fixed capital and share of industry and services to GDP. Within a mentioned sector, identification and effectively address of binding constraints in a particular sector is also termed as structural change.

$$d_t = \gamma_0 + \sum_{k=1}^n \gamma_k \sin\left(\frac{2\pi kt}{T}\right) + \sum_{k=1}^n \delta_k \cos\left(\frac{2\pi kt}{T}\right), n \leq \frac{T}{2} \quad (5)$$

If the estimate has n cumulative frequencies, then k represents a specific frequency, and T shows the number of observations. In this situation, we don't observe any nonlinear trend for all values of $\gamma_k = \delta_k = 0$. As a result, the specification proposed by Leybourne et al. (1998b) becomes a special case. The use of a large value of n is not recommended for various quality reasons and this phenomenon will be demonstrated. However, the main problem is that it can result in an over-fitting problem. Numerous illustrious studies, such as Gallant (1981), Davies (1977), Gallant and Souza (1991), and Bierens (1997), empirically show that one can frequently capture the essential characteristics of an unknown functional form smooth break by using a small number of frequency variables and the Fourier approximation. Additionally, since a gradual progression of the nonlinear trend is necessary, n should be low. The test equation can finally be provided in the form below:

$$\Delta \delta_t = \gamma_0 + \sum_{k=1}^n \gamma_k \sin\left(\frac{2\pi kt}{T}\right) + \sum_{k=1}^n \delta_k \cos\left(\frac{2\pi kt}{T}\right) + \phi_1 \varepsilon_{t-1} + \sum_{k=1}^n \phi_k \Delta \delta_{t-1} + u_t \quad (6)$$

To test the stationary dynamics of $\hat{\varepsilon}_t$ by using Eq. (6), it is standard practice to increase the lag value of the dependent variables. The question of whether a limited number of frequency elements, may mimic the kinds of breaks that are typically seen in economic data is a crucial one to consider in Kim and Perron's unit root test. To tackle this issue, we used a single frequency element started with a Fourier flexible approximation k breaks, and then evaluated the amplitude and displacement of the sinusoidal element of the deterministic term by γ_k and δ_k . In this manner, we can allow various smooth breaks with a single frequency $k = 1$.

H_0 : Unit Root (Linear non-stationary).

H_1 : Nonlinear stationary (Nonlinear and stationary around smooth breaks).

We will use critical values of smooth structural break unit root test to test the hypothesis against the critical values.

4.2. Bootstrapping ARDL bounds testing approach to cointegration

To assess the relationship between the variables, we consider the bootstrapping ARDL (BARDL) cointegration technique that was developed by McNown et al. (2018). The key advantage of this methodology is that it resolves the issue of poor size and power attributes which was presented in the conventional ARDL method first proposed by Pesaran and Shin (1999). This method supports the inclusion of a new test and complements the conventional ARDL limits testing approach. It also increases the power of the F -test and t -test. If Pesaran et al. (2001) ARDL cointegration technique is not used, three tests are therefore necessary to confirm the existence of the cointegration relationship between the variables. Two requirements must be met for Pesaran et al. (2001) to establish a cointegration relationship: both the coefficients of the lagged explanatory variables and the error-correction coefficients must be statistically significant. The first condition, however, is applicable only if the lagged dependent variable is statistically significant in terms of error correction; similarly, the second situation is only applicable if the lagged explanatory factors are statistically significant. The critical bounds, such as upper and lower bounds, proposed by Pesaran et al. (2001), are applied in the second instance, though there are no bound tests or critical bounds in the first. When all of the sample variables are stationary at first order, that is, integrated of order 1, these tests can be applied in the first scenario where the error-correction coefficients are statistically significant. However, a crucial point to remember in this situation is that due to their poor explanatory and power features,

conventional unit root tests may be troublesome (Goh et al., 2017). The bootstrapping ARDL (BARDL) test of McNown et al. (2018) model can address this issue by employing new statistics. As shown by Monte Carlo simulations, the critical values of the bootstrapping model have a larger size and power properties.

$$\tilde{C}_t = \sum_{i=1}^p \alpha_i \tilde{C}_{t-i} + \sum_{r=0}^q \beta_r \tilde{X}_{t-r} + \sum_{z=0}^r \alpha_z \tilde{Z}_{t-z} + \sum_{h=0}^s \gamma_1 \tilde{D}_{t,h} + \mu_t \quad (7)$$

where i, r, z and h stands the lags ($i=1, 2, \dots, p$; $r=0, 1, 2, \dots, q$; $z=0, 1, 2, \dots, r$; $h=0, 1, 2, \dots, s$ and t represent the time), $\tilde{C}_t$ is the response variable in the model, and $\tilde{X}_t$ and $\tilde{Z}_t$ are the explanatory variables. $\tilde{D}_{t,h}$ is the dummy variable, β and α represent the coefficients of lagged explanatory variables, and γ is the coefficient of dummy variable. μ_t represents the error-term with zero mean and finite variance. Based on this model, we have written in an error-correction equation as follows:

$$\Delta C_t = \phi C_{t-1} + \gamma X_{t-1} + \psi Z_{t-1} + \sum_{i=1}^{p-1} \alpha_i \Delta C_{t-i} + \sum_{r=1}^{q-1} \beta_r \Delta X_{t-r} + \sum_{z=1}^{r-1} \alpha_z \Delta Z_{t-z} + \sum_{h=1}^s \gamma_1 D_{t,h} + \varepsilon_t \quad (8)$$

$\phi = \sum_{i=1}^p \alpha_i$, $\gamma = \sum_{r=0}^q \beta_r$ and $\psi = \sum_{z=0}^r \alpha_z$, at this point, α_i , β_r , α_z , and γ_1 stand for the function related with Eq. (8). Taking vector auto-regression levels into its error-correction equation, the derivation of Eq. (8) from Eq. (7) is estimated. Whereas Eq. (7) can be estimated by using the constant term (C) in the unconditional model that can be specified as:

$$\Delta C_t = \bar{\gamma} + \phi \bar{C}_{t-1} + \bar{\gamma} X_{t-1} + \bar{\psi} Z_{t-1} + \sum_{i=1}^{p-1} \bar{\alpha}_i \Delta C_{t-i} + \sum_{r=1}^{q-1} \bar{\beta}_r \Delta X_{t-r} + \sum_{z=1}^{r-1} \bar{\alpha}_z \Delta Z_{t-z} + \sum_{h=1}^s \bar{\gamma}_1 D_{t,h} + \bar{\varepsilon}_t \quad (9)$$

To verify the cointegration between the variables C_t , X_t and Z_t , all three null hypotheses need to be rejected.

The hypothesis can be as follows:

- F1 test based on all relevant terms of error correction ($H_0: \phi = \gamma = \psi = 0$ against H_1 : any of ϕ , γ , ψ are different from zero).
- F2 test based on all of the explanatory variable's terms ($H_0: \gamma = \psi = 0$ against H_1 : either γ or ψ is different from zero).
- t -test based on the lagged dependent variable ($H_0: \phi = 0$ against H_1 : ϕ is different from zero).

It is important to note that the conventional ARDL approach only generates the critical values of the bounds test for $F1$ and t -tests. $F2$ test results for the lagged explanatory variables are omitted, nevertheless. Critical values for all three tests can be obtained by employing the bootstrapping strategy ARDL recommended by McNown et al. (2018). At the same time, to give empirically sound results, we employed the critical values tabulated by McNown et al. (2018) in this research.

4.3. Granger causality approach based on bootstrapping ARDL

The conventional Granger causality tests will be used to determine the direction of the long-run and short-run causal relationships. The Granger causality test is a useful tool for providing insights into the nature and direction of relationships between the variables (Tashevska et al., 2020). The Granger causality approach based on Bootstrapping has been successfully applied in the research in econometrics by multiple studies (Bayar et al., 2020; Payab et al., 2023; Siddiqui et al., 2022). In the absence of cointegration between y and x when y is dependent variable, the Granger causality test for $x \geq y$ should include the lagged differences on x only, implying that, we test whether $\delta = 0$. On the other hand, if there exists cointegration between the variables, it implies that the dependent and the independent variables form a stationary linear combination. As a result, the lagged levels can be treated as $I(0)$. In this

case, Granger-causality test for $x \geq y$ should include the lagged differences on x and the lagged level of x , i.e. test whether $\beta > 0$ and $\delta = 0$. We can also extend Eq. (10) to three-variable case (see the following model):

$$\Delta \tilde{y}_t = \alpha + \beta x_{t-1} + \gamma \tilde{z}_{t-1} + \sum_{i=1}^{p-1} \theta_i \Delta \tilde{y}_{t-i} + \sum_{i=1}^{p-1} \delta_i \Delta \tilde{x}_{t-i} + \sum_{i=1}^{p-1} \omega_i \Delta \tilde{x}_{t-i} + \sum_{j=1}^q \eta \tilde{D}_{t,j} + \mu_t \quad (10)$$

In this case, the Granger-causality test for $\tilde{x} \geq \tilde{y}$ should include the lagged differences on $\tilde{x}$ and lagged level of $\tilde{x}$, to test whether $\beta > 0$ and $\delta = 0$. For $\tilde{z} \geq \tilde{y}$ should include the lagged differences on z and the lagged level of z , to test whether $\delta > 0$ and $\omega = 0$ (if they are cointegrated). Cointegration testing with the Bootstrap ARDL method has a number of intriguing features. First, compared to other multivariate cointegration techniques, it performs better on small samples. Second, the inclusion of both I(0) and I(1) time series (but not I(2)) in a long-run relationship is made possible by the fact that it does not impose the restrictive condition that all series are integrated of the same order. The latter allows for flexibility and prevents potential “pre-test bias” which refers to the long-run model specification using only I(1) variables (Pesaran et al., 2001).

5. Empirical results and their discussion

5.1. Descriptive analysis and pair-wise correlations

The empirical results of descriptive statistics and correlation analysis are shown in Table 1. We find that variations in energy diversification are lower than variations that occur in natural capital. Financial development is more volatile compared to volatility stems in economic growth, exports diversification and structural changes. The results of Jarque-Bera test show that energy diversification, financial development, natural capital, structural changes, economic growth and export diversification have normal distribution. This leads us for further empirical results. The correlation analysis reveals the presence of positive correlation between energy diversification and financial development. Natural capital is positively correlated with energy diversification. The positive correlation exists between structural changes and energy diversification. Economic growth and export diversification is negatively correlated with energy diversification. Structural change and export diversification (economic growth) is inversely (positively) correlated with financial development. A negative correlation exists between natural capital and structural change. Exports diversification and economic growth have negative correlation with natural capital. Economic growth is negatively correlated with structural change. The positive correlation is found between exports diversification and economic growth. Exports diversification and economic growth have a positive correlation.

Table 1
Descriptive Statistics and Correlation Analysis.

Variable	$\ln E_t$	$\ln F_t$	$\ln N_t$	$\ln S_t$	$\ln Y_t$	$\ln X_t$
Mean	0.2688	2.8039	1.0198	0.7857	2.7805	-0.1202
Median	0.2768	2.8412	1.0210	0.7820	2.7914	-0.1159
Maximum	0.2848	2.9024	1.0243	0.8177	2.8190	-0.0757
Minimum	0.2435	2.6225	1.0136	0.7644	2.7131	-0.1536
Std. Dev.	0.0147	0.0874	0.0032	0.0164	0.0308	0.0244
Skewness	-0.3854	-0.6448	-0.6810	0.4125	-0.6611	0.1549
Kurtosis	1.4399	2.1226	2.0336	1.7648	2.2391	1.6705
Jarque-Bera	0.1362	0.1949	0.1255	0.9928	1.0474	0.83853
Probability	0.8710	0.8141	0.8918	0.6911	0.5093	0.7015
$\ln E_t$	1.0000					
$\ln F_t$	0.0603	1.0000				
$\ln N_t$	0.1786	0.0119	1.0000			
$\ln S_t$	0.1002	-0.1765	-0.5146	1.0000		
$\ln Y_t$	-0.2757	0.4712	0.1028	-0.1092	1.0000	
$\ln X_t$	-0.1566	-0.4144	0.2410	0.0578	0.2807	1.0000

5.2. Unit root analysis

For testing the stationary level of all the variables, we have applied the traditional ADF unit root test and ADF unit root test with a single unknown structural break. The empirical results of both unit root tests are reported in Table 2. The empirical results reported in Table 2 show that energy diversification, financial development, natural capital, structural changes, economic growth and export diversification are found non-stationary at levels containing intercept and time trend. Further, we find that all the variables show an absence of unit root problem at 1st difference. This shows that energy diversification, financial development, natural capital, structural changes, economic growth and export diversification are integrated at I(1). It is indicated by Shahbaz et al. (2018) that the traditional ADF unit root test may provide ambiguous empirical results due to ignoring the role of structural break (s) stemming in the time series data. This ambiguity of empirical results by unit root test such as the traditional ADF unit root test may mislead us in choosing an appropriate cointegration approach for examining cointegration relationship between the variables. We chose to apply Kim and Perron (2009) ADF unit root test contains information of single unknown structural break in the series for overcoming the above issue. The results of the structural break ADF unit root test is also presented in Table 2. Following empirical results reported by the structural break ADF (Kim and Perron, 2009) unit root test, we find that all the variables

Table 2
Unit Root Analysis.

Variables	ADF Unit Root Test		ADF Unit Root Test with Single Break		
	T-Statistic	Prob. Value	T-Statistic	Prob. Value	Break Year
$\ln E_t$	-2.3687(2)	0.3934	-4.6359(2)	0.1792	2009 _{QI}
$\ln F_t$	-3.0372(1)	0.1298	-3.2365(4)	0.9140	2012 _{QII}
$\ln N_t$	-1.5739(4)	0.7963	-4.8148(3)	0.1199	2014 _{QIV}
$\ln S_t$	-1.9192(5)	0.6318	-4.6336(2)	0.1800	2009 _{QI}
$\ln Y_t$	-1.7460(4)	0.7235	-4.2597(5)	0.3589	2006 _{QI}
$\ln X_t$	-2.1546(2)	0.5090	-4.0321(3)	0.4090	1998 _{QIII}
	-6.2670(3)	*	-5.3762(3)	*	
$\Delta \ln E_t$	*	0.0000	**	0.0290	1997 _{QI}
	-4.3565(2)	*	-9.3365(4)	*	
$\Delta \ln F_t$	*	0.0036	*	0.0010	2008 _{QI}
	-4.6603(4)	*			
$\Delta \ln N_t$	*	0.0014	5.9793(3)*	0.0010	1998 _{QI}
	-4.3394(4)	*	-5.6810(1)	*	
$\Delta \ln S_t$	*	0.0041	**	0.0118	1998 _{QIV}
	-5.9484(3)	*	-6.5540(2)	*	
$\Delta \ln Y_t$	*	0.0000	*	0.0010	2019 _{QIV}
	-4.2860(2)	*	-6.7661(3)	*	
$\Delta \ln X_t$	*	0.0049	*	0.0010	1998 _{QI}

Note: * and ** show significance at 1% and 5% levels of significance respectively. () indicates appropriate lag order chosen based on AIC.

contain unit root problem at level with intercept and time trend in the presence of single structural break stems in the series. These structural breaks are found in the series of energy diversification, financial development, natural capital, structural changes, economic growth and export diversification over the years of 2009Q_I, 2012Q_{II}, 2012Q_{IV}, 2009Q_I, 2006Q_I and 1998Q_{III}. The presence of structural breaks in the series may be an outcome of energy reforms, financial reforms, reforms linked with natural resources (capital), structural reforms, economic reforms and trade reforms regarding exports implemented in Australian economy over the period of 1995–2021.

Energy diversification contains a structural break in 2009Q_I which is an indication of the Australian economy's initiative(s) in the energy sector. Basically, Australia has been undergoing a rapid energy transition by diversifying its energy infrastructure since 1990. There have been several factors that have led to energy diversification in the country such as technological change (e.g. low carbon technologies), decarbonization, and innovation (BREE, 2013). This profound transformation in the energy sector aimed to design the energy market to tackle climate issues. Under the Paris agreement, Australia has ambitious plans to set greenhouse gas emissions reduction targets by encouraging efficient technologies and renewable energy sources. Moreover, a reduction in reliance on coal-based energy over the years (i.e. 12% for the period of 2006–2016) has helped reduce emissions. There has been a significant increase in energy demand during the 2000s, which spurred energy production to meet both domestic and international energy demand. Yet, Australia is heavily relying on coal, gas, and oil energy sources. In 1988, the first formal energy policy discussion paper was released by the Australian government, which aimed to find strategies for creating an enabling environment to promote clean energy production and consumption in the country (Department of Defence, 2009, 2013). There were subsequent working papers, which focused on the future fuel strategy for greater consumer choices and emission reductions. The Australian government is committed to encouraging the use of renewable energy sources by attracting clean energy investment. Fig. 1 shows the trends of renewable energy consumption in Australia from 1973 to 2015. On average, less than 10% of total energy consumption is from renewable(s). However, the share of the energy mix (particularly, renewables) varies from state to state. In the last two decades, the Australian government has increased its spending on energy-related research significantly (IEA, 2018). For instance, from 2004 to 2013, the government's spending on research & development (R&D) increased many fold (i.e. by AU\$961 million).

After 1st differencing, we note that energy diversification, financial development, natural capital, structural changes, economic growth, and export diversification are found stationary by containing information of a single structural break in the series. This confirms that all the variables are integrated at I(1) in the presence of a single structural break stems in

the series. We may conclude that the results reported by Kim and Perron (2009) show the robustness of our empirical analysis regarding the investigation of the unit root problem of the variables.

5.3. Bootstrapped ARDL cointegration analysis

After establishing that all variables are integrated at order I(1), the next step is to investigate the cointegration relationship between energy diversification and its drivers (1). For this, we used bootstrapping autoregressive distributive lagged modelling (BARDL) to determine whether the variables are in long run cointegration equilibrium. Shahbaz et al. (2018) document that the BARDL bounds testing method performs better than the conventional ARDL model. Regarding cointegration equilibrium between the sample variables, this method considers the joint F-test on all lagged level variables, the *t*-test on the lagged level of the dependent variable as well as a new *t*-test on the lagged level of the regressors. This demonstrates that for evaluating cointegration, the bootstrapping ARDL test is superior to the straightforward ARDL limits testing method. When using the BARDL bounds testing approach to ascertain cointegration between the variables, lag length selection is required. The incorrect choice of lag length results in misleading empirical estimates. Due to its superior power qualities, we find Akaike Information Criteria (AIC) to be more appropriate for variables with the right lag order (Lütkepohl, 2006). In column 2 of Table 3, the variables' chosen lag lengths are displayed.

In the backdrop of bootstrapping ARDL, we computed the *t*-value and *F*-value to examine the long-run cointegration connection among the variables, and empirical results are presented in Table 3. The null hypothesis, which treats energy diversification, financial development, natural capital, structural changes, and economic growth as dependent variables, is rejected by these bootstrapping statistics, such as the *t*-test and *F*-test, both of which take into account the lagged level of dependent variable. Furthermore, the alternative hypothesis is similarly accepted by the *t*-test on the lagged explanatory factors. Given that we considered export diversification as a dependent variable, we are unable to reject the null hypothesis. In the demand function for energy diversification, we discover the existence of four cointegrating vectors. This further implies that the presence of a long-run equilibrium cointegration connection between variables at the 1% and 5% levels, respectively, is confirmed by the joint *F*-test, *t*-test on the lagged dependent variable, and *t*-test on the lagged independent variable.

5.4. Long run regression analysis

The presence of cointegration further leads us to examine the long run impact of financial development, natural capital, structural changes, economic growth, and exports diversification on energy diversification.

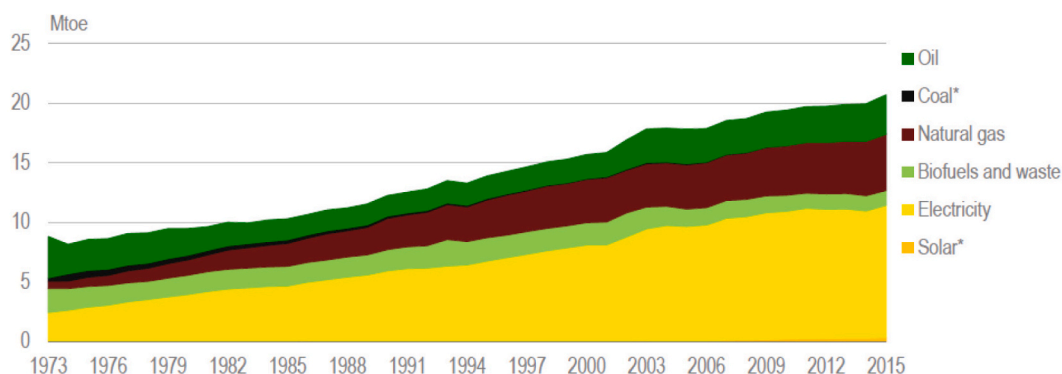


Fig. 1. Australia's Electricity Diversification.

Source: International Energy Agency, 2017, www.iea.org/statistics.

Table 3
Bootstrapped ARDL Cointegration Analysis.

Bootstrapped ARDL Cointegration Approach						Diagnostic Tests			
Estimated Models	Lag Length	Break Year	F _{PSS}	T _{DV}	T _{IV}	$\bar{R}^2$	Q – stat	LM(2)	JB
$E_t = f(F_t, N_t, S_t, Y_t, X_t)$	6, 5, 5, 6, 5, 5	2009 _{QI}	9.1021*	−3.9020**	−4.7101*	0.7931	3.8010	1.7016	0.9022
$F_t = f(E_t, N_t, S_t, Y_t, X_t)$	6, 5, 5, 6, 5, 6	2012 _{QII}	8.1809*	−4.8050*	−4.6010*	0.6034	5.5089	2.0103	0.7080
$N_t = f(E_t, F_t, S_t, Y_t, X_t)$	6, 5, 5, 6, 5, 6	2014 _{QIV}	7.9169*	−5.2087*	−3.0515**	0.7952	6.1640	2.2252	0.7606
$S_t = f(E_t, F_t, N_t, Y_t, X_t)$	6, 5, 6, 5, 5, 6	2009 _{QI}	9.6511*	−5.2022	−5.6910*	0.2911	5.7010	0.4506	0.8073
$Y_t = f(E_t, F_t, N_t, S_t, X_t)$	6, 5, 5, 5, 5, 5	2006 _{QI}	7.1091	−3.9292**	−4.1096*	0.2820	5.1126	0.5306	0.8565
$X_t = f(E_t, F_t, N_t, S_t, Y_t)$	6, 6, 5, 5, 5, 6	1998 _{QIII}	1.2202	−0.9010	−1.6803	0.6051	4.6060	1.9050	0.8908

Note: Significant values of 1% and 5% are indicated by the asterisks * and **, respectively. The optimal lag length is determined by the Akaike Information Criterion (AIC). The bootstrap approach produces F-statistic known as FPSS, which is based on the asymptotic critical bounds. Lagrange Multiplier test (LM) and Jarque-Bera test (JB) are both t-statistics. TDV is t-statistic for the dependent variable and TIV is t-statistic for the independent variables.

Table 4
Long Run Regression Analysis.

Dependent Variable: $\ln E_t$				
Variable	Coefficient	T-Statistic	Coefficient	T-Statistic
Constant	−0.0603	−0.1289	−5.9667*	−10.5057
$\ln F_t$	0.1550**	2.9048	5.8854*	12.7194
$\ln F_t^2$			−1.0128*	−12.417
$\ln N_t$	0.7658*	4.7925	0.2093***	1.8987
$\ln S_t$	0.0295**	2.2543	0.2198*	2.9352
$\ln Y_t$	−0.3842**	−2.3148	−0.9621*	−8.4021
$\ln X_t$	−0.3552*	−8.0351	−0.3561**	2.8476
D_{2009QI}	−0.0106*	−5.8663	−0.0059*	−4.9156
R^2	0.9380		0.9756	
Adjusted- R^2	0.9343		0.9739	
F-statistic	25.4890*		50.3321*	
Prob.	0.0000		0.0000	
D-W stat ^a	2.2418		2.2363	

Stability Analysis				
Test	F-statistic	Prob.	F-statistic	Prob.
χ^2_{NORMAL}	0.7678	0.7061	0.3361	0.8161
χ^2_{ARCH}	1.3060	0.2170	0.1568	0.4512
χ^2_{WHITE}	0.2802	0.8706	0.1209	0.9090
χ^2_{SERIAL}	0.1030	0.7573	0.2516	0.6540
χ^2_{RESET}	0.9080	0.2105	0.4987	0.4870
CUSUM	Stable		Stable	
CUSUMsq	Stable		Stable	

Note: *, ** and *** show significant at 1%, 5% and 10% levels respectively.

^a Durbin-Watson.

The long run results are reported in Table 4. We find that financial development has a positive and statistically significant effect on energy diversification. By keeping all else is same, a 1% increase in financial development leads energy diversification by 0.1550% at 5% level of significance. Over the past years, the Australian government has reinvigorated the development finance policies to promote clean energy by creating enabling environment for innovative finance development to reduce emissions. The relationship between natural capital and energy diversification is positive and statistically significant. We note that natural capital has a positive effect on energy diversification. Other things remain the same; a 1% increase in natural capital stimulates energy diversification by 0.2093% to 0.7658%. In recent years, Australian government has established the Clean Energy Finance Corporation (CEFC). The government gave the CEFC AUD 10 billion to invest in the clean energy industry and to help attract more funding from other investors. Structural changes have a positive effect on energy diversification, which is statistically significant. It is noted that a 1% change in structural reforms (changes) may lead the change in energy diversification by 0.0295% to 0.2198%, keeping other factors similar. In late 2011, the government introduced policy reforms to drive the transformation to a clean energy economy in the long-run. If Australia is to reduce its greenhouse gas emissions in accordance with its international

commitments, there needs to be a far higher investment in new clean technologies and solutions. Economic growth is negatively linked with energy diversification, and it is statistically significant. All being same, a 1% increase in economic growth reduces energy diversification by 0.3842% to 0.9621%. One of the reasons that until 2010s, there was a little focus on renewable energy investment by the Australian government, prior to that period, most of the energy production and trade concentrated on coal. The association between export diversification and energy diversification is negative and statistically significant. Export diversification declines energy diversification. Keeping other things, the same, a 1% increase in export diversification lowers energy diversification by 0.3552% to 0.3561%. The Australian economy is largely dependent on energy exports to the rest of world. As a notable energy exporter, in the year 2012–2013, it exported about 80% of its energy produced from different sources (such as coal, oil, gas, and uranium) dominated by coal energy (Azad et al., 2015). Australia has begun to invest in alternative energy sources, but more funding still has to be set aside for the implementation of renewable energy technologies. Dummy variable has a negative but significant effect on energy diversification. This indicates that the Australian government's initiatives started for energy diversification could not produce fruitful results. This further shows that the reforms initiated in or energy sector especially implemented in 2009 have negative effect on energy diversification. Though Australia has started taking climate change policy action in early 2000s, the energy diversification and transition did not get momentum until recently. Energy exports majorly have been confined to coal exports, which may be one of the reasons that dummy variable shows a negative impact on energy diversification.

To examine non-linear association between financial development and energy diversification, we have included squared term of financial development in energy diversification demand function. The empirical results are reported in Table 4. The linear and squared terms of financial development show the presence of inverted-U relationship between financial development and energy diversification. We find that a 1% increase in financial development increases energy diversification by 5.8854% but negative sing of nonlinear (squared) terms seems to corroborate the delinking of energy diversification and financial development at higher levels of financial development. This reveals that energy diversification increases initially with financial development and declines after threshold level of financial development. These findings are in line with economic theory as energy diversification requires a significant initial capital investment (particularly, renewable energy). If we look at financial development trajectory of Australia, initially, billions of dollars were spent to boost clean energy infrastructure.

5.5. Short run regression analysis

The results of short run analysis are reported in Table 5. We find that financial development has negative but an insignificant effect on energy diversification. The relationship between natural capital and energy diversification is negative and it is statistically significant at 1% level of

Table 5
Short Run Regression Analysis.

Dependent Variable: $\Delta \ln E_t$				
Variable	Coefficient	Std. Error	T-Statistic	Prob.
Constant	−0.0002	0.0002	−0.9095	0.3653
$\Delta \ln F_t$	−0.0479	0.0365	−1.3110	0.1929
$\Delta \ln N_t$	−0.3730*	0.1297	−2.8760	0.0049
$\Delta \ln S_t$	−0.0165	0.0551	−0.2999	0.7648
$\Delta \ln Y_t$	0.2039**	0.0987	2.0667	0.0414
$\Delta \ln X_t$	0.0825**	0.0382	2.1564	0.0335
D_{2009Q_3}	−0.0006*	0.0001	−3.4511	0.0008
ECM_{t-1}	−0.0860*	0.0187	−4.5940	0.0000
R^2	0.4136			
Adj- R^2	0.3722			
F-statistic	9.9197*			
Prob.	0.0000			
D-W stat ^a	2.0351			

Stability Analysis		
Test	F-statistic	Prob.
χ^2_{NORMAL}	0.1910	0.9217
χ^2_{ARCH}	0.1818	0.6707
χ^2_{WHITE}	0.1276	0.8712
χ^2_{SERIAL}	0.1517	0.8248
χ^2_{RESET}	1.1196	0.2656
CUSUM	Stable	
CUSUMsq	Stable	

^a Durbin-Watson.

significance. Structural changes have negative but insignificant effect on energy diversification. Economic growth is positively and significantly linked with energy diversification. Export diversification has a positive and significant effect on energy diversification. The effect of dummy variable on energy diversification is negative and it is statistically significant at 1% level of significance. The estimate of ECM_{t-1} (lagged error term) is negative and significant at 1% significance level. The statistically significance with a negative sign of lagged error term is further proof for the existence of long run between the variables. The coefficient of ECM_{t-1} is -0.0860 which entails that variations occur in energy diversification from short run to long run are corrected by 8.60% in each quarter which will take 2 years and 10 months to reach an equilibrium path. The diagnostic analysis reveals the normal distribution of residual term. We note that evidence of no serial correlation, no autoregressive conditional heteroskedasticity and no White heteroskedasticity is found. The Ramsey Reset test analysis confirms the well-specification of short run model. The stability of long run and short run analysis is tested by applying CUSUM and CUSUMsq tests. We find that graphs of CUSUM and CUSUMSQ lie within critical boundary at 5% level of significance (Fig. 2a, Fig. 2b). This corroborates the stability of long-run and short-run parameters.

5.6. Bootstrapped ARDL based Granger causality analysis

The empirical results of the Bootstrapped ARDL based causality approach are reported in Table 6. In the long run, we find that financial

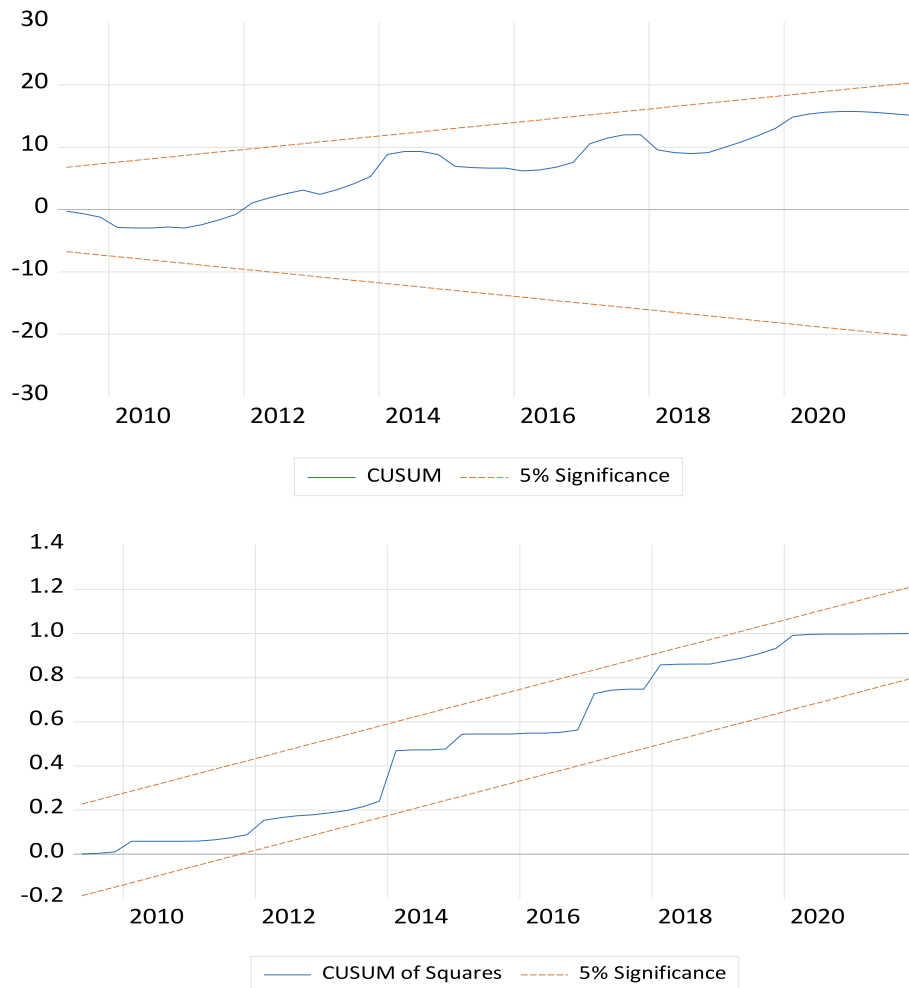


Fig. 2. a: CUSUM.
b: CUSUMsq.

Table 6

BARDL Based Long-Run and Short-Run Causality Analysis.

Dependent Variable	Direction of Causality							Long Run
	Short Run						Break Year	
	$\Delta \ln E_{t-1}$	$\Delta \ln F_{t-1}$	$\Delta \ln N_{t-1}$	$\Delta \ln S_{t-1}$	$\Delta \ln Y_{t-1}$	$\Delta \ln X_{t-1}$		
$\Delta \ln E_t$		2.4043(+)** [0.0375]	1.2915(-) [0.2798]	0.1740(+) [0.8405]	1.4080(-) [0.2998]	0.4021(-) [0.6700]	2009 _{QI}	-0.0678* [-3.8172]
$\Delta \ln F_t$	4.8890(+)** [0.0004]		0.1134(+) [0.9821]	8.8933(+)* [0.0003]	0.0041(-) [0.9959]	6.3309(+)* [0.0000]	2012 _{QII}	-0.0395** [-2.0879]
$\Delta \ln N_t$	2.1106(-) [0.1270]	0.0534(-) [0.9557]		7.5463(+)* [0.0000]	1.0026(-) [0.3709]	2.1854(-) [0.1182]	2014 _{QIV}	-0.0544** [-2.4469]
$\Delta \ln S_t$	0.0687(+) [0.9336]	4.9412(-)* [0.0092]	5.2699(-)* [0.0004]		0.1947(-) [0.8234]	2.2962(-) [0.1069]	2009 _{QI}	-0.0138* [-4.1079]
$\Delta \ln Y_t$	1.8228(+) [0.1674]	0.5138(+) [0.59999]	0.8976(-) [0.3727]	1.2640(-) [0.2874]		5.7611(-)* [0.0044]	2006 _{QI}	-0.0870** [-2.9328]
$\Delta \ln X_t$	0.5509(+) [0.5783]	3.1496(-)** [0.0501]	1.5907(+) [0.2093]	3.0495(+)** [0.0522]	5.1405(-)* [0.0076]		1998 _{QIII}	-0.0543 [-1.0105]

Note: * and ** show significance at 1% and 5% levels of significance.

development Granger causes energy diversification and as a result, energy diversification Granger causes financial development. This confirms the presence of a feedback effect between financial development and energy diversification. These results imply that while energy diversification favours financial development to support sustainable economic growth, financial development supports the shift to alternate green energy sources to reduce emissions. This is consistent with Australian government plans that aim to implement low-carbon energy technology in order to attain net-zero goals. It is noted that financial development positively causes energy diversification and similar causality is also found running from energy diversification to financial development. The causal association between natural capital and energy diversification is bidirectional. This shows that natural capital has positive causation towards energy diversification and energy diversification causes natural capital in Granger sense. These results are consistent with the Australian government's commitment to increase knowledge of natural capital investment at the national and regional levels in order to reduce the impact of climate change on the productivity that drives economic activity. Similarly, a feedback effect i.e. bidirectional causality exists between structural changes and energy diversification. It is noted that structural changes Granger cause energy diversification positively and as a result, energy diversification positively causes structural changes. The Austrian economy's growth strategies have been supported by structural reforms throughout the past three decades, which were mostly market-driven improvements that led to energy diversification. The diversity in energy mix with a focus on renewables is aimed to accelerate productivity and achieving the net-zero goals. The relationship between economic growth and energy diversification is bidirectional i.e. negative feedback effect. This shows that economic growth negatively Granger causes energy diversification and similarly, energy diversification causes economic growth negatively in Granger's sense. The postulation of feedback hypothesis implies a mutual relationship between energy diversification and economic growth, which suggests that the institutionalization of energy and economic policies meant to reduce such factors could be counterproductive in defiance of both energy security and economic wellbeing. The negative bidirectional causality also noted between energy diversification and export diversification. One of the possible reasons of this negative causal relationship may be that Australian exports (largely dependent on resources and agricultural sectors) have been concentrated in China and East Asian countries. The export concentration may be the result of market forces particularly, the resources sector's exports (such as coal and iron), which are majorly destined to China.

In the short run, a positive feedback effect is noted between financial development and energy diversification. The neutral effect is noted between natural capital and energy diversification. Neither structural changes cause energy diversification nor energy diversification causes

structural changes in Granger's sense. Economic growth negatively causes energy diversification and as a result, energy diversification positively causes economic growth in Granger sense but it is statistically insignificant. Similarly, energy diversification positively causes export diversification insignificantly and export diversification negatively causes energy diversity insignificantly. The energy diversification strategy by the Australian government since 2012 has resulted in closure a majority of its coal fired power plants (Goddard and Farrelly, 2018). This effort of energy diversification and shift towards alternative energy sources has resulted in energy production reduction of almost 50% of present capacity (Li et al., 2020). In light of this fact, it is imperative that Australia must reduce its dependence of carbon-intensive fuels and diversify its energy generation sources to minimise the impact on climate change.

6. Conclusion and policy implications

This study investigated the association between financial development and energy diversification by including natural capital, structural changes, economic growth and export diversification into energy diversification demand function for Australian economy over the period of 1995–2021. For empirical purpose, we applied advanced ADF unit root test developed by Kim and Perron (2009) which contains information of unknown single structural break for examining unit root properties of the variables. The bootstrapping ARDL cointegration approach applied for investigating cointegration relationship between financial development, natural capital, structural changes, economic growth, export diversification and energy diversification by considering presence of structural break stems in the series. The Granger causality approach based on Bootstrapping ARDL also applied for examining the short-run and long-run causal association between energy diversification and its determinants.

Our empirical indicated that all the variables are showed that at first difference, all variables were found to be stationary. The Bootstrapping ARDL bounds testing's analysis also shows that energy diversification and its determinants have cointegration for long run relationship for the period of 1995–2021 in the presence of structural break in the series. Empirical results also showed that financial development has favourable impact on energy diversification. The significant and positive relationship exists between natural capital and with energy diversification. Structural changes are positively linked with energy diversification. Economic growth is negative linked with energy diversification. Export diversification has a negative effect on energy diversification. The nonlinear relationship between financial development and energy diversification is inverted-U shaped. The empirical analysis of Granger causality approach based on Bootstrapping ARDL reveals that financial development and energy diversification have bidirectional causal

relationship. The feedback effect exists between natural capital and energy diversification. The bidirectional causality exists between structural changes and energy diversification. The relationship between economic growth and energy diversification is bidirectional i.e. negative feedback effect. The negative bidirectional causality is also found between energy diversification and export diversification.

Our empirical findings offer profound implications macro and microeconomic policies to address product and factor markets by guiding the government, investors, and regulators to improve financial development and environmental management through well-developed infrastructure and institutional capacity. The Australian energy diversification policies could enable the country in achieving net zero targets through energy efficiency and performance by providing demand side solutions to reduce energy cost pressures, lessening emissions, and delivering a high performing economy by achieving the sustainable development goals of climate change and environmental sustainability. On other hand, the export market diversification could help to ensure the demand by shifting towards alternative sources of energy (e.g. hydrogen, and renewables). Capital formation has a strong and positive relationship with green energy. For instance, climate capital investment initiatives in Australia offer opportunities to unlock renewable energy sources. With a specific focus on alternative energy sources like hydrogen, Australia is emerging as a global leader in developing hydrogen industries alongside China, Japan, Korea, and other significant trading partners. However, to meet its future domestic clean energy demand, electrifying other sectors of the economy (such as transport and heavy industry) currently dependent on fossil fuels, and exporting clean energy to the rest of the world it would require breaking new ground for low-cost capital investment.

Policy makers needs to focus on increasing scope of financial development to provide clean energy through solutions through diversified energy source. By collaborating with international partners, the government should concentrate on creating an enabling environment that is favourable to investment. The government may try to attract a sizable amount of foreign investment in clean technology in order to turn Australia into a “clean energy giant” and diversify its energy sources. The government’s climate change-related initiatives have also stimulated investment in huge renewable electricity generating. Increased investment has been made possible by developers of renewable generation power plants having better access to financing. In addition to boosting the local economy and employment, the establishment of new industries like clean hydrogen would open up new export markets and job opportunities and help other countries reduce their emissions more quickly. Australia must collaborate and co-invest with other nations in order to speed up technological advancement and decarbonize the entire world economy, not just our own. The scalability of global manufacturing and supply networks will make it less expensive to deploy the technologies that all countries, including Australia, need. Australia must therefore build on its achievements, enhance the competitiveness of its economic sectors, and further use its natural resources to create more jobs and expand its market share in both traditional and low-emissions markets, employment and economic opportunities with low emissions along with traditional markets.

As the globe transitions to a cleaner global economy, conventional, new energy sources and mineral resources (including copper, cobalt, and lithium) are in high demand. Australia has an unparalleled quantity of these resources. New export markets and employment opportunities are being produced by the rising industries. Exploiting the commodities and new minerals (such as include uranium and lithium), which permit reduced emissions for batteries, that will see rising worldwide demand in the upcoming years due to both new and existing businesses will help strengthen the economy and create jobs. The Australian government must prioritise enabling infrastructure for the diversification of clean energy exports to maintain improvement and refinement in its policies for reducing emissions.

Investing in R&D and a technology-led approach would not only

assist Australians avoid paying a bigger proportion of their household bills for energy but will also lower the cost of new and developing technologies. The initiatives for business collaboration and support could assist in implementing innovative technologies to fulfil the rising demand for low-emissions energy and goods and so increase shared economic gains. The government must have a clear role in ensuring that Australia accomplishes its carbon reduction targets, that the transition away from coal-fired electricity is orderly, maintains energy security, prevents price spikes, helps employees make the switch, and does so in a way that avoids price spikes. In order to fulfil expanding global demand, supply channels for essential minerals must be strong and durable, which necessitates diversifying these supply chains. Thanks to the energy transition, Australia has a once-in-a-generation opportunity to capitalise on the increasing demand for low-emission technologies and increase our capacity to create value at home. The development of a well-defined energy policy framework to support clean energy investment infrastructure will support the domestic consumption and energy export that could fetch significant benefits growing and strengthening Australian economy.

Since, financial development and energy diversification is an emerging topic and dynamic in nature which needs further attentions to develop this nexus based on theoretical backgrounds by applying recent empirical approaches. For example, financial development-energy diversification nexus can be extended by adding new and potential determinates of energy diversification such as urbanization, trade diversification, currency devaluation etc. All regional blocks such G 7 countries, E 7 countries and D 7 countries have been focusing to diversify their energy sources in the presence of emerging financial sector. In doing so, recent panel approaches considering cross-sectional dependence and structural breaks in data are available in existing literature to examine the cointegration between the variables and to check the impact of financial development, financial depth or financial inclusion (including other potential factors) on energy diversification. In applications pertaining to the time series, there is a vital significance of quantile information. One of the primary benefits of the quantile regression is that they present an approach for estimation of the conditional quantiles. The second advantage is that they extend the option for modelling by permitting to use local as well as quantile-specific analysis of time series. The greatest disadvantage of simple conditional mean method is that they may not be able to reveal some critical facts about the association between the random variables. This is so, as many research questions may be extremely complex and hence the simple conditional methods may not provide complete explanation. Therefore, for the purpose of obtaining better understanding of association between random variables, quantile regression approaches are more suitable.

CRedit authorship contribution statement

Muhammad Shahbaz: Conceptualization, Formal analysis, Writing – original draft, Supervision. **Aaliyah Siddiqui:** Conceptualization, Methodology, Writing – original draft. **Shabbir Ahmad:** Methodology, Software, Writing – original draft. **Zhilun Jiao:** Conceptualization, Writing – review & editing, Project administration.

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Appendix A. Supplementary data

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